

Supplementary Material: Tactile Simulation and Dexterous-Hand Experiments

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A.1 Scope of the Supplementary Material

This supplementary material expands implementation and experimental details that are only summarized in the main paper. Section A.2 focuses on tactile simulation, including MuJoCo tactile readouts, pressure reconstruction, and multi-finger tactile graph representation learning, and further describes how the four-finger temporal tactile representation is trained from these data. Section A.3 supplements the experimental protocol for cross-hand mapping and the task suite on the real Allegro and Inspire dexterous hands. Section A.4 analyzes tactile contribution, cross-hand transfer, and generalization.

A.2 Tactile Simulation and Four-Finger Temporal Tactile Graph Representation

A.2.1 MuJoCo Tactile-Sensor Simulator

Simulator goal. We build a MuJoCo-based tactile-sensor simulator to model the ADC readings of the fingertip tactile patches used in this work. The simulator uses the hand pose, object pose, and contact geometry provided by MuJoCo to compute fingertip contact location, contact area, and local pressure, and then converts these physical contact quantities into ADC responses with a customized resistive taxel model. Unlike directly using simulated contact forces, this module explicitly models the taxel layout, pressure diffusion, noise, and crosstalk of the tactile patch, making the simulated readings closer to real sensor outputs. Depending on the experiment, the system outputs either 12-dimensional single-finger ADC readings or 48-dimensional four-finger ADC readings, which are further used for pressure reconstruction, contact recognition, and four-finger temporal tactile representation learning. The simulator mainly complements real demonstrations with contact states that are difficult to cover systematically, such as weak contact, changes in contact area, different grasp poses, and early slip, and therefore provides a repeatable data source for tactile readout modeling and tactile representation learning.

A.2.2 Sensor and ADC Modeling

Each fingertip tactile patch contains 4×3 taxels, giving 12 sensing points per finger. The single-finger setting uses a 12-dimensional ADC vector. In the four-finger Allegro setting, index, middle, ring, and thumb readings are concatenated into a 4×12 ADC array and flattened into a 48-dimensional ADC vector. ADC values are in the range 0–4000. Before tactile-encoder training, the readings are normalized within each sensing region to reduce scale differences across contact strengths and fingertips. Dense pressure maps use a 48×32 resolution and are used to evaluate whether low-dimensional taxel readings preserve contact location, total force, and pressure distribution. Table 3 summarizes the sensor layout and ADC settings used throughout the tactile experiments.

Table 3: Tactile sensor and ADC settings.

Item	Setting
Single-finger taxels	$4 \times 3 = 12$
Four-finger input	4×12 ADC array, flattened as a 48-dimensional vector
ADC range	0–4000
Dense pressure map	48×32
Patch size in unfolded coordinates	30 mm $\times$ 45 mm
Sensor model	Resistive voltage-divider ADC with noise and crosstalk

A.2.3 Single-Finger and Curved-Fingertip Calibration

Single-finger calibration consists of planar scanning and curved-fingertip mapping. In the planar stage, a rounded probe presses the tactile patch at multiple force levels, and the 12 taxel ADC readings are recorded as input for dense pressure-map reconstruction. The test contact shapes include point, two points, disk, bottle-cap ring, bar, and ellipse, covering spatial structures from localized point contact to extended surface contact.

The same tactile patch is then mounted onto a curved Allegro fingertip. Calibration data are recorded in an unfolded sensor coordinate system, allowing the planar pressure-reconstruction procedure to be reused in the curved setting. The reconstructed pressure map is then visualized on the curved fingertip geometry. This experiment tests whether ADC[12] still represents contact center and total pressure under realistic fingertip curvature and mounting direction. The scan scale is listed in Table 4, and the corresponding planar and curved reconstruction examples are shown in Figure 6.

Table 4: Curved-fingertip calibration scale.

Item	Value
Samples	7656
Scan grid	29×44
Force levels	0.2, 0.5, 1.0, 2.0, 3.0, 4.0 N
Scan step	1.0 mm
Boundary margin	0.75 mm
Probe diffusion sigma	2.0 mm

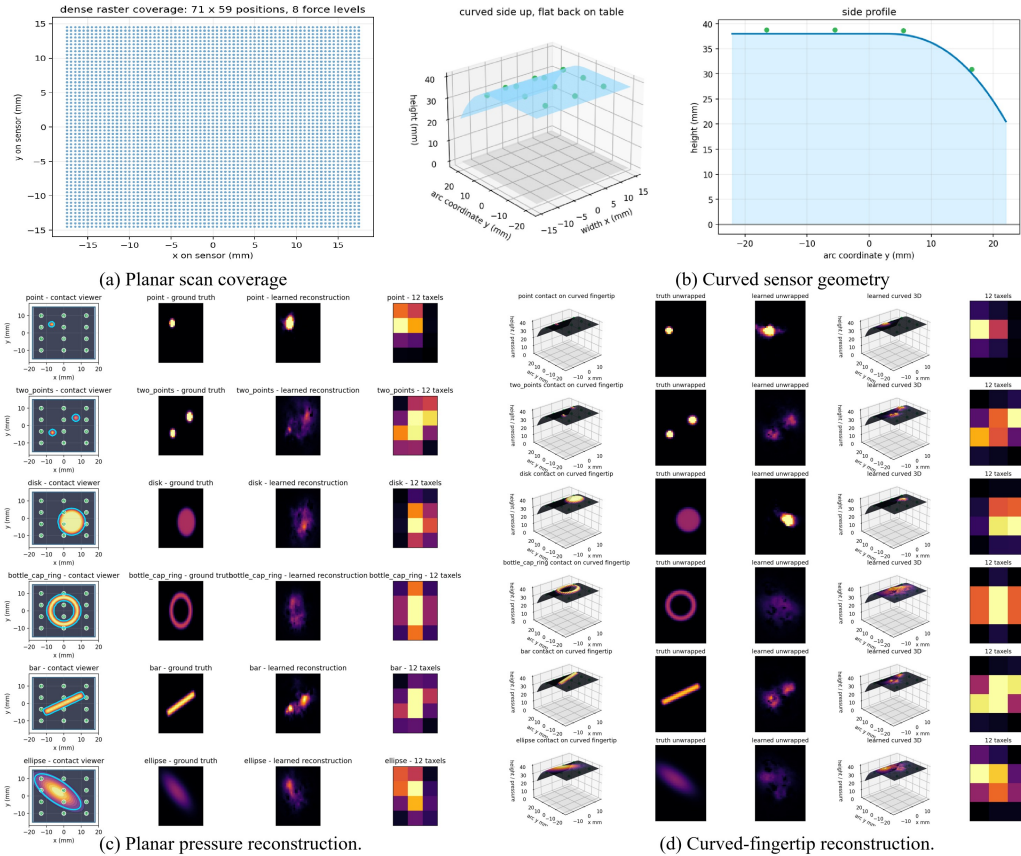


Figure 6: Single-finger and curved-fingertip tactile calibration. The simulator checks whether low-dimensional ADC readings can recover spatial pressure structure in both planar and curved mounting configurations.

Under a 4 N load, curved-fingertip reconstruction predicts total force close to the ground truth, and the contact-centroid error remains at the millimeter scale. These results indicate that the 12-taxel response preserves the main contact-location and pressure-distribution information despite its low dimensionality. Table 5 reports representative reconstruction errors for the tested contact shapes.

Table 5: Representative curved-fingertip reconstruction errors under a 4 N load.

Shape	True force/N	Pred. force/N	Force error/N	Centroid error/mm
point	4.000	4.371	0.371	1.067
two points	4.000	3.997	-0.003	1.788
disk	4.000	3.995	-0.005	0.888
bottle cap ring	4.000	4.084	0.084	0.139
bar	4.000	4.409	0.409	0.449
ellipse	4.000	3.923	-0.077	1.428

A.2.4 Four-Finger Grasp Tactile Input

The multi-finger setting extends tactile sensing to four Allegro fingertips. The object set contains cube, rectangular box, cylinder, sphere, ellipsoid, and bottle. Each sample stores four fingertip readings as ADC[4,12], which are flattened into ADC[48]. The classification experiment is used only as a diagnostic task: it checks whether four-finger tactile signatures contain object-shape and contact-pattern information, rather than serving as the final policy objective. The data scale and split are summarized in Table 6, and the diagnostic pressure maps and ADC signatures are visualized in Figure 7.

Table 6: Four-finger tactile classification data.

Item	Value
Object classes	6
Variants per class	10
Repeats per variant	20
Total samples	1200
Train/val/test	720 / 240 / 240
Input	4×12 ADC array flattened as a 48-dimensional vector
Best validation accuracy	0.9958
Test accuracy	0.9875

The classifier using the 48-dimensional ADC vector reaches 0.9875 test accuracy. This diagnostic result suggests that four-finger tactile input is not limited to a binary contact signal; it also captures pressure distribution, contact shape, and grasp-configuration differences. These properties motivate using ADC[48] as the input for the four-finger temporal tactile representation in Section A.2.5.

A.2.5 Four-Finger Temporal Tactile Graph Representation

The four-finger temporal tactile module corresponds to the frozen tactile encoder in the main paper. It learns a frozen tactile encoder f_{tac} before policy training, so the VLA action expert receives a stable contact feature instead of learning low-level pressure geometry from limited real trajectories. Each 48-dimensional ADC frame is represented as a 48-node tactile graph. Dense edges within each fingertip model local pressure structure, while weaker cross-finger edges connect the thumb to the other fingers to encode coordinated multi-finger contact. A graph encoder maps each frame to a frame-level tactile token, and a temporal Transformer aggregates tokens over time into the final tactile representation.

In implementation, an ADC sequence with shape $[B, T, 48]$ is converted into a tactile graph sequence with shape $[B, T, 48, 9]$. The graph encoder processes each frame and produces frame-level tactile tokens, while the temporal Transformer aggregates these tokens across time to obtain the final tactile representation f_{tac} with shape $[B, D]$. The temporal data are generated by the tactile simulator and synchronized with Allegro joint states, object pose, and contact state. The pretraining objectives include masked pressure reconstruction, next-frame prediction, contact-state prediction, sequence-summary prediction, and full pressure-sequence reconstruction. Masked reconstruction checks whether the encoder preserves local pressure distributions. Next-frame and contact prediction constrain short-term contact transitions. Summary prediction encourages the final tactile representation to retain statistics of the whole contact sequence. Figure 8 shows masked pressure-reconstruction examples, and Table 7 summarizes the pretraining objectives.

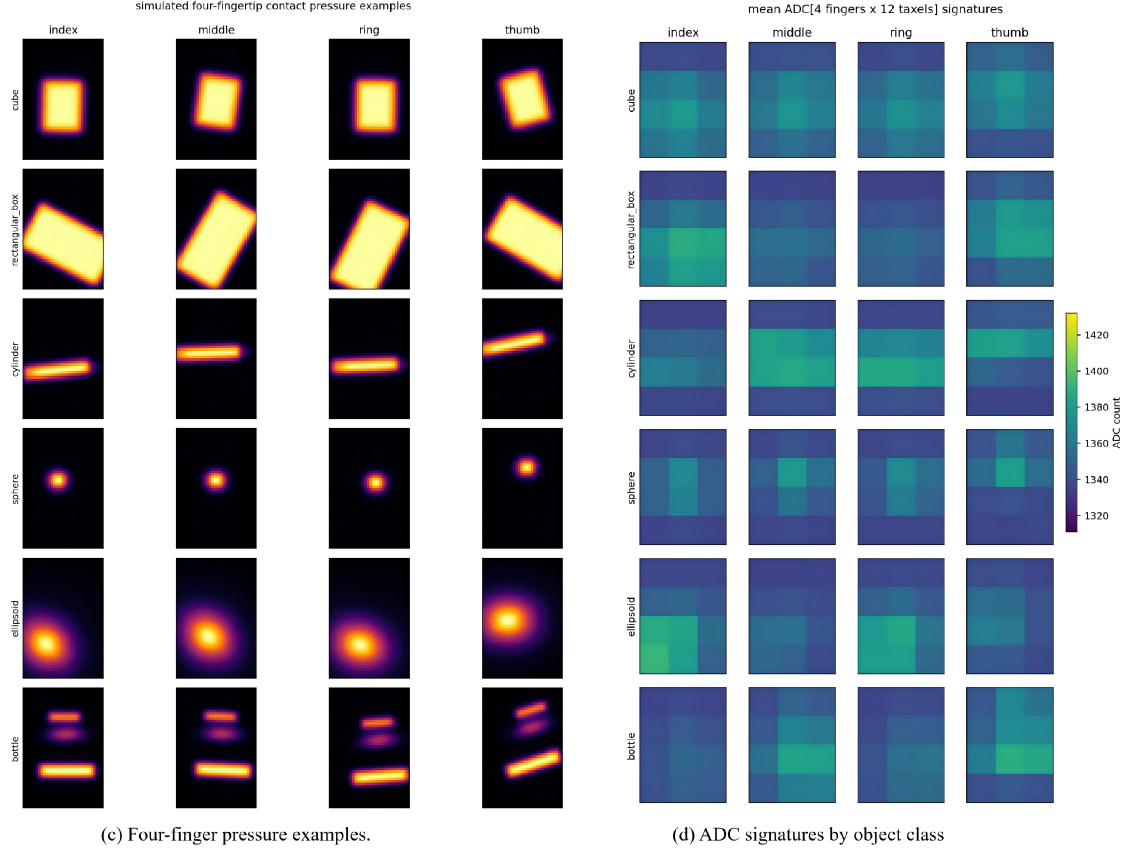


Figure 7: Four-finger tactile diagnostics. The 48-dimensional ADC vector carries more than binary contact information; it reflects pressure distribution, grasp configuration, and object-contact geometry.

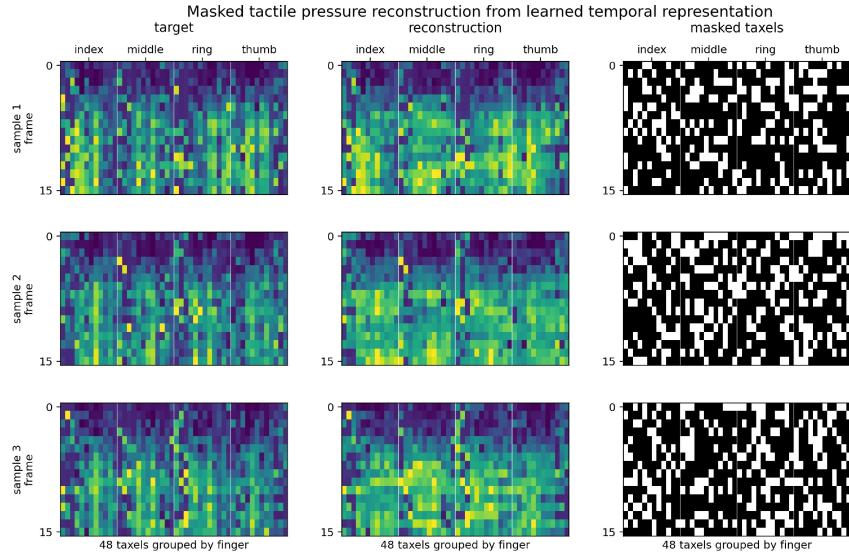


Figure 8: Four-finger temporal tactile reconstruction examples. The masked pressure reconstruction task checks whether the frozen tactile encoder preserves sequence-level pressure structure across the four fingertips.

Table 7: Temporal tactile pretraining objectives.

Objective	Meaning
L_mask	Reconstruct randomly masked taxel pressure values
L_next	Predict the next pressure frame from previous tactile tokens
L_contact	Predict contact state, per-finger contact, total pressure, and contact count
L_summary	Predict sequence-level statistics from f_{tac}
L_ftac_recon	Reconstruct the full pressure sequence from the final tactile representation

During VLA training, f_{tac} is used as a frozen tactile feature for the action expert. This separates tactile geometry and pressure-pattern learning from policy optimization, keeps the tactile interface stable during cross-hand transfer, and supports controlled ablations of tactile fusion, tactile gating, and hand-specific adapters.

A.3 Real Dexterous-Hand Experiment Protocol

Figure 9 supplements the real-system experiments with representative Inspire-hand task examples. The real platform consists of a UR5e arm, replaceable dexterous hands, a tabletop manipulation workspace, a fixed external camera, and a wrist camera. The figure shows the Inspire hand mounted at the UR5e end effector and executing real manipulation tasks, including the target objects, tabletop workspace, and task space observed by the cameras. Real demonstration data are collected using GELLO and WiseGlove. GELLO is used to teleoperate the UR5e arm and provides end-effector poses and arm motion trajectories. WiseGlove records human hand motion, which is mapped through hand-specific adaptation to drive the dexterous hand. Each trajectory synchronously records language instructions, visual observations, proprioceptive states, demonstrated actions, contact states, shared-interface labels, and target-hand commands. These data are used to train and evaluate the shared contact-kinematic action interface and the target-hand adapter/decoder.

In cross-hand experiments, the UR5e arm, table, camera placement, workspace, object set, and language templates are kept fixed. The only switched components are the dexterous-hand hardware and the corresponding target-hand adapter/decoder. This design keeps the comparison focused on the effectiveness of the shared action interface and target-hand decoding, rather than on environment or task-initialization differences.



Figure 9: Real Inspire-hand experiment rollouts. The figure shows representative task executions for pick-and-place, wiping, box opening, and cutting under the same tabletop setup used in the cross-hand evaluation.

A.4 Generalization, Data Ablation, and Tactile Contribution

Section A.4 adds three diagnostic experiments to analyze the sources behind the averaged success rate. First, we report cross-hand transfer separately for each target hand: Allegro $\rightarrow$ Inspire is evaluated on the real system, whereas Allegro $\rightarrow$ Leap is evaluated only in simulation because a real Leap hand is not available. This split helps reveal transfer differences across target hands instead of relying only on a pooled average. Second, we compare Sim-only and Real+sim data settings to study the role of simulated data in the performance change. Third, we attach the tactile-fusion module to ACT and DP / Diffusion Policy backbones to examine whether tactile information consistently improves policies with different action-modeling assumptions.

Table 10: Transfer success rates.

Setting	SR(%)
Allegro $\rightarrow$ Inspire	88.1
Allegro $\rightarrow$ Leap	87.5

Table 11: Data ablation.

Data setting	SR(%)
Sim-only	56.3
Real+sim	88.1

Table 12: Tactile-fusion comparison.

Backbone	No tactile	+ tactile
ACT	38.8	48.75
DP / Diffusion Policy	51.9	56.3

Cross-Hand Transfer. Table 10 reports Allegro $\rightarrow$ Inspire and Allegro $\rightarrow$ Leap separately. The former is evaluated on the real system, whereas the latter is evaluated only in simulation due to the absence of a real Leap hand. Reporting the two settings separately makes the transfer condition explicit and avoids hiding target-hand differences behind a single average.

Real and Simulated Data. Table 11 shows that Real+sim outperforms Sim-only. This result indicates that simulated data does not directly replace real demonstrations, but can serve as an effective complement to them. Its main role is to expand the coverage of contact states, object poses, and interaction processes, thereby improving stability in contact-rich tasks. We therefore position simulated data as contact-coverage augmentation for real demonstrations, rather than as an independent substitute for real data.

Tactile Fusion Across Backbones. Table 12 compares the effect of tactile fusion at the policy-backbone level. With ACT, tactile fusion improves the success rate from 38.8% to 48.75%, corresponding to a 9.95 percentage-point increase. With DP / Diffusion Policy, the success rate improves from 51.9% to 56.3%, corresponding to a 4.4 percentage-point increase. The improvement on both backbones suggests that the tactile encoder provides contact information that is useful beyond a single policy architecture.